

Human Motion Generation from Text, Scene, and Trajectory on NymeriaPlus

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Abstract

Human motion generation conditioned jointly on text, scene, and trajectory remains underexplored, largely because few real-world datasets pair motion with both scene context and language annotations at scale. We address this by training on NymeriaPlus, a large-scale dataset of real-world egocentric motion capture with paired scene and text annotations. We propose a diffusion model in which text, scene, and a goal position are encoded as tokens, concatenated onto the motion sequence, processed together by a single self-attention transformer, and rolled out autoregressively to generate long motions. Each condition is dropped independently during training in the style of classifier-free guidance, so at sampling time any subset of the three can be supplied, omitted, or guided at its own strength. Qualitative rollouts on held-out test sequences show motion that follows the text, respects the scene, and reaches the goal, and a per-condition guidance ablation shows that each guidance weight controls its own condition.

1. Introduction

Realistic human motion is needed in many fields, from character animation in games and film to virtual and augmented reality, synthetic training data, and embodied agents and robots that interact with people. In all of these settings, the motion must look natural, follow a text instruction, respect the surrounding scene, and reach a specified goal location. Most prior work addresses only a subset of these conditions, and the methods that combine all of them are trained on synthetic scenes, since few real-world datasets pair motion with both scene and language annotations.

We address this with a motion diffusion model that treats text, scene, and goal in the same way: each is turned into a small set of tokens and appended to the motion sequence, so that a single transformer reasons over the motion and all conditions jointly. During training, each condition is dropped independently, in the style of classifier-free guidance (CFG) [3]. At inference, the model can therefore generate from any subset of the three conditions and be steered

toward each of them with its own guidance scale.

Unlike prior scene-aware methods, we train on real-world capture rather than synthetic scenes, using NymeriaPlus [1], a large-scale dataset of egocentric motion with paired scene and text annotations. Long sequences are generated autoregressively, one window at a time. We validate our approach on held-out sequences, showing that the generated motion respects the text, the scene constraints, and the specified trajectory, and that each guidance scale controls its own condition.

2. Related Work

Motion diffusion models. Diffusion models generate data by learning to reverse a gradual noising process: a network is trained to recover a clean motion sequence from a noised version of it, and at sampling time it is applied iteratively, starting from pure noise. MotionDiffuse [12] was among the first to apply this framework to text-driven human motion, predicting the added noise with a transformer that attends to text features. MDM [10] instead predicts the clean motion directly, which allows geometric losses and constraints to be applied at every step. Both are trained on text-annotated motion capture such as HumanML3D [2], which contains no scene information. We follow the MDM formulation and extend it with scene and goal conditioning.

Scene-aware motion generation. Placing a person in a 3D environment requires motion data paired with the surrounding scene, which is much harder to capture than motion alone. Existing datasets therefore rely on synthetic components. HUMANISE [11] builds language-conditioned human-scene interactions by placing captured motion into scanned indoor scenes. TRUMANS [5] records motion in a large set of synthetic scenes and generates long interaction sequences autoregressively, conditioned on the scene and a target trajectory. LINGO [4] extends this setting with text instructions and is the closest to our work, as it supports text, scene, and trajectory together. In all three cases, however, either the scenes or the pairing between motion and scene is synthetic. To our knowledge, NymeriaPlus is the only dataset that pairs real, in-the-wild motion with



Figure 1. Qualitative rollouts on held-out test sequences. Each row shows eight evenly spaced frames (left to right) from one continuous rollout, with its text prompt below.

both text and scene annotations, and it is also by far the largest (Tab. 1). This makes it possible to learn all three conditions from real data, which is the setting we study.

3. Method

Our model generates motion in short windows, each expressed in a local coordinate frame and conditioned on text, scene, and goal. We first describe the data and the motion representation, then the architecture and the multi-condition guidance, and finally how long motions are generated win-

dow by window.

3.1. Dataset

We use NymeriaPlus [1], which extends Nymeria [7]. Nymeria is a large-scale multimodal dataset of real-world daily activities, recorded with a motion-capture suit and multiple egocentric wearable devices and annotated with hierarchical text descriptions. NymeriaPlus adds improved body motion in SMPL [6] format, together with 3D annotations and reconstructions of indoor objects and structural

Table 1. Methods and datasets for conditioned motion generation, in chronological order, compared by supported conditions, data source, and hours of motion.

Method / Dataset	Data	Text	Scene	Traj.	Hours
HumanML3D [2]	Real	✓	–	–	28.6
HUMANISE [11]	Synthetic	✓	✓	–	11
MDM [10]	Real	✓	–	–	–
TRUMANS [5]	Synthetic	–	✓	✓	15
LINGO [4]	Synthetic	✓	✓	✓	16
NymeriaPlus [1]	Real	✓	✓	✓	182

elements, which define the scene. We keep only sequences with complete SMPL tracking, object geometry, and narrations, and discard annotations shorter than 16 frames or longer than 60 frames. The resulting corpus contains 707 sequences, 181.96 hours of motion at 10 FPS, and 144,356 atomic text annotations.

We split the data so that each participant and each room belongs to exactly one split. This yields 548 training, 83 validation, and 76 test sequences, and no test room or participant is seen during training.

3.2. Motion Representation

Each frame is represented by the 3D positions of the 24 SMPL joints, flattened into a 72-dimensional vector. We work at 10 FPS, and the model operates on windows of 16 history frames followed by 16 frames to be generated.

All quantities in a window, namely the motion, the goal, and the scene crop, are expressed in a local coordinate frame, the *anchor*, defined at the last history frame (or at the first frame when there is no history). Its axes are aligned with the body’s facing direction, and its origin lies directly below the pelvis, at the floor height estimated from foot contacts.

3.3. Architecture

Following MDM [10], we use a transformer denoiser that predicts the clean motion directly from a noised window, with a cosine noise schedule and 50 diffusion steps. The transformer has 8 layers, a model dimension of 512, 8 attention heads, and 1024-dimensional feed-forward blocks. The diffusion timestep modulates every block through AdaLN-Zero [8] rather than entering the token sequence. Following TRUMANS [5], each window is conditioned on the history frames that precede it, which enables autoregressive generation of long sequences (Sec. 3.5). These history frames are kept fixed at every denoising step, and the training loss is a mean squared error on joint positions over the generated frames only.

Shared self-attention. As shown in Fig. 2, the tokens of every active condition are concatenated onto the motion tokens, and each block applies a single self-attention layer

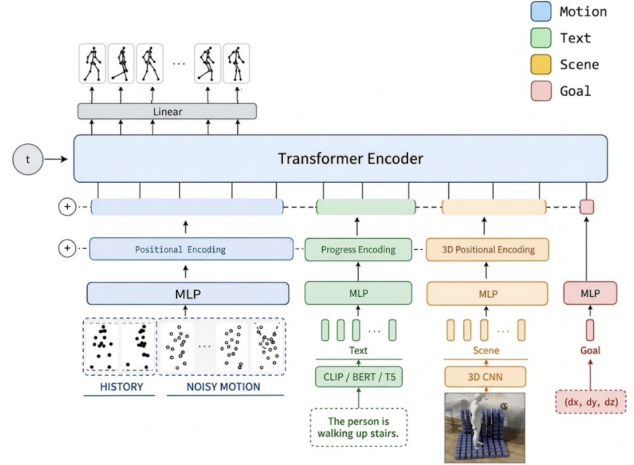


Figure 2. Architecture overview. Motion, text, scene, and goal tokens are concatenated into one sequence and processed by a single transformer denoiser.

to the whole sequence instead of a separate cross-attention path per condition. A learned token-type embedding for each condition, together with a sinusoidal positional encoding on the motion tokens, lets attention distinguish history from future frames and one condition from another.

Text. Text is encoded with a frozen T5-Small [9] encoder and kept as a sequence of per-token embeddings, padded or truncated to 64 tokens, rather than pooled into a single vector. This lets individual phrases of a prompt align with different parts of the generated window. Since an annotation usually spans several windows that all share the same text, we also embed a window-progress scalar, which indicates where the window falls within the annotation, and add it to the text tokens.

Scene. The scene of each sequence is voxelized into a binary occupancy grid at 5 cm resolution. As shown in Fig. 3, we crop a 3.2 m cube around the anchor and encode it with a small 3D CNN down to a fixed $4 \times 4 \times 4$ volume, yielding 64 scene tokens. Each token receives a 3D positional encoding, computed by passing the anchor-local coordinates of its cell center through a small MLP, so that attention knows where each token lies in space.

Goal. The goal is the target pelvis position at the end of the window, expressed as an anchor-local displacement (dx, dy, dz) . The arrival time is implied by the window length, so no explicit timing is needed. Like text and scene, the goal enters the attention sequence as a single token rather than being hard-inpainted into the generated window. The model is therefore *encouraged* to reach it but can trade it off against what the scene or text imply.

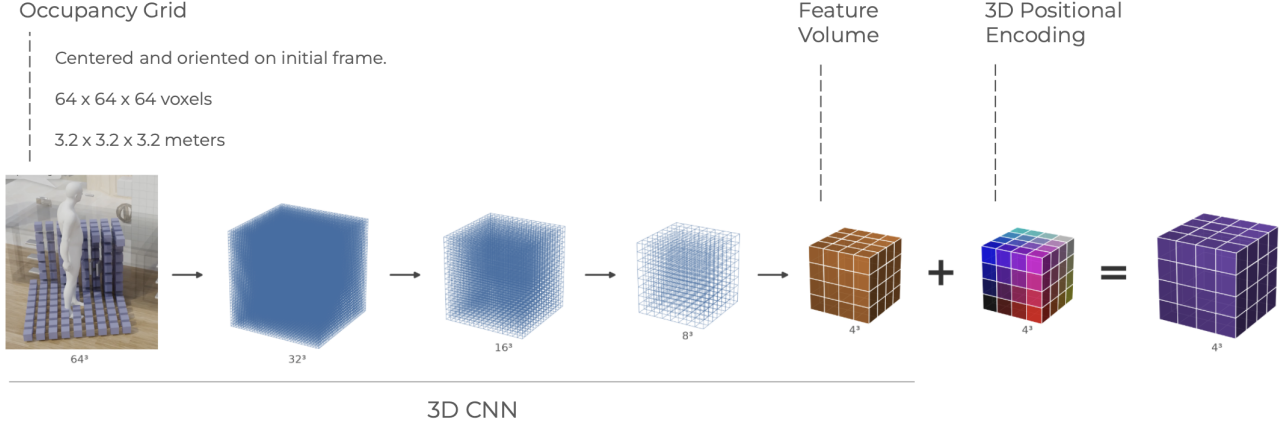


Figure 3. Scene encoder. A 3D CNN reduces the occupancy crop around the anchor to a $4 \times 4 \times 4$ feature volume and adds a 3D positional encoding.

3.4. Multi-Condition Classifier-Free Guidance

We extend classifier-free guidance [3] to multiple conditions. During training, each of text, scene, and goal is dropped independently with probability 20%, and all three are dropped together with an additional probability of 5%. A dropped condition simply omits its tokens from the sequence, so no learned null embedding is needed.

At sampling time, let $\hat{\mathbf{x}}(T, S, G)$ denote the model’s clean-motion prediction with text, scene, and goal tokens T, S, G all present, and let $\emptyset$ indicate that a condition is dropped. Each condition defines its own *guidance direction*, the change in the prediction when only that condition is removed:

$$\Delta T = \hat{\mathbf{x}}(T, S, G) - \hat{\mathbf{x}}(\emptyset, S, G), \quad (1)$$

$$\Delta S = \hat{\mathbf{x}}(T, S, G) - \hat{\mathbf{x}}(T, \emptyset, G), \quad (2)$$

$$\Delta G = \hat{\mathbf{x}}(T, S, G) - \hat{\mathbf{x}}(T, S, \emptyset). \quad (3)$$

The guided prediction adds each direction to the fully conditional prediction, scaled by that condition’s guidance weight w_c :

$$\hat{\mathbf{x}}_{\text{guided}} = \hat{\mathbf{x}}(T, S, G) + \sum_{c \in \{T, S, G\}} (w_c - 1) \Delta c. \quad (4)$$

3.5. Autoregressive Rollout

At inference, long motions are generated one window at a time (Fig. 4). The first window starts without history, anchored at a given start position on the floor. For every following window, the 16 most recent generated frames are passed to the model as history, so each new window continues the previous motion instead of starting from scratch.

Before each new window, the anchor is recentered on the body: its horizontal position moves to the pelvis of the last

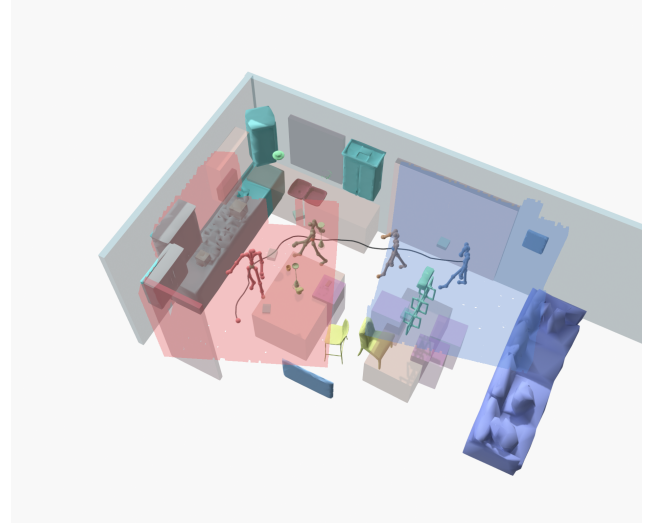


Figure 4. Autoregressive rollout. The last frames of each window become the history of the next, and the anchor is recentered before every window.

generated frame, its facing direction is re-estimated from the hip joints, and its floor height is re-estimated from the history and the newly generated frames. The history, the goal, and the scene crop are then re-expressed in this new local frame. Every window is therefore seen in the same body-centered, floor-anchored coordinates as during training, which lets the rollout continue for an arbitrary length and move freely through the scene.

4. Qualitative Results

We evaluate the model qualitatively on the 76 held-out test sequences (Sec. 3.1), whose rooms and participants were

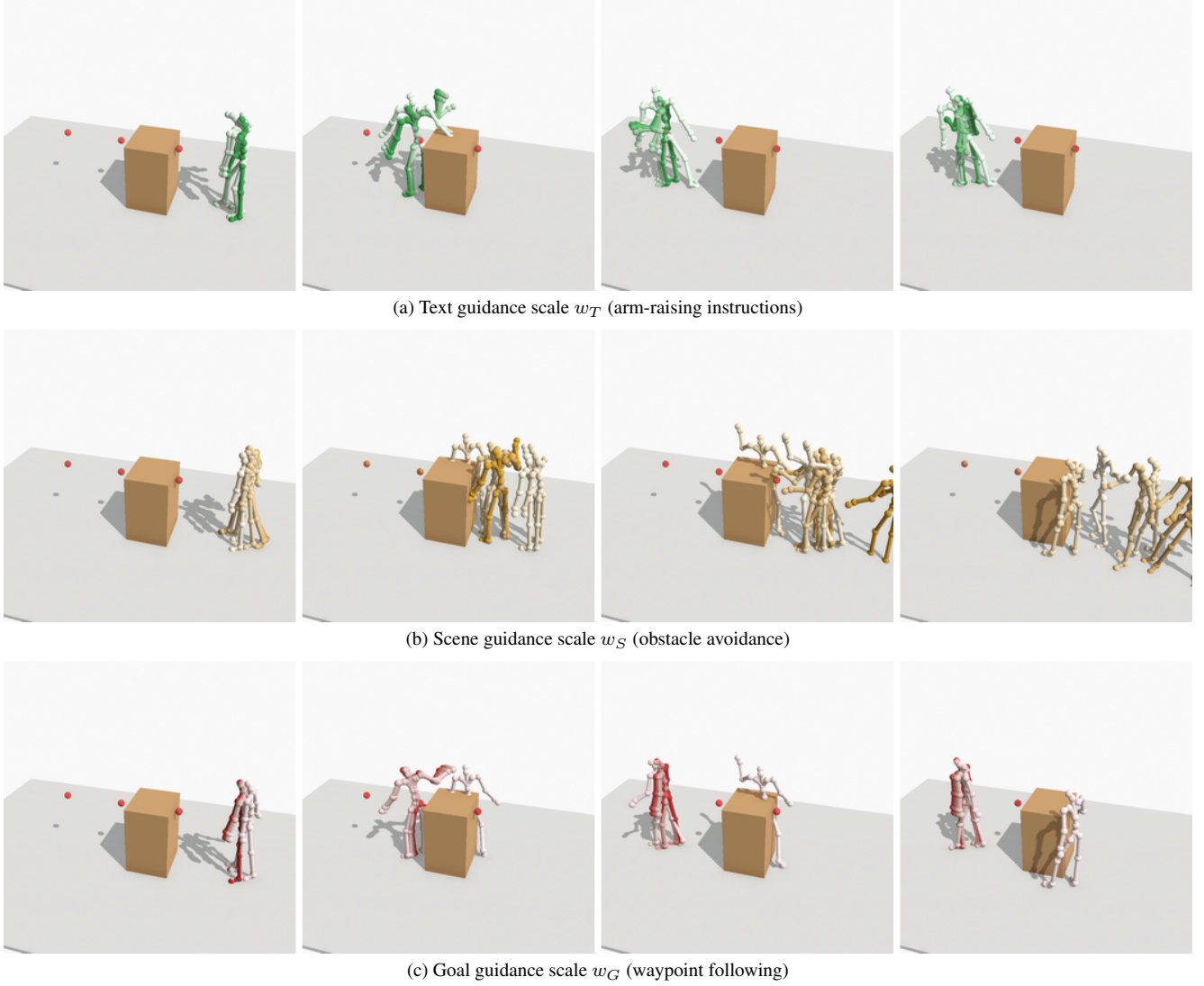


Figure 5. CFG ablation. Prompts: “The person walks forward and raises their right hand.” then “The person lowers their right hand and raises their left hand.” Red spheres mark the waypoints. In each row, one guidance scale is varied from 0 to 3 (left to right) while the other two are fixed at 1. Color intensity encodes the guidance scale: the higher the scale, the stronger the color.

never seen during training. All rollouts are conditioned jointly on text, scene, and goal. We produce them with an interactive tool in which a user places a start position and a sequence of segments directly on the scene, each with a text prompt and a waypoint, and adjusts the guidance scale of each condition.

Figure 1 shows that a single model covers a wide range of everyday behaviors. The person walks naturally through the room and transitions smoothly into the action described by the text, such as stirring food at the counter, opening the fridge with both hands, or lying down on the bed. The same model also handles large changes in height, from climbing stairs to sitting on a sofa, and keeps the body clear of the surrounding furniture throughout. In each case, the motion

stays consistent with the text, the scene, and the goal at the same time, rather than falling back to a generic walking cycle.

4.1. CFG Ablation

To examine what each guidance weight of Sec. 3.4 controls, we generate a two-segment rollout in a simple synthetic scene with a single box-shaped obstacle, and sweep the guidance scale of one condition at a time while keeping the other two fixed (Fig. 5).

Each weight affects its own aspect of the motion. Increasing the text guidance makes the described arm raises more pronounced, increasing the scene guidance keeps the trajectory clear of the obstacle, and increasing the goal guid-

ance moves the rollout further toward the waypoints. Because the three scales are independent, the balance between the conditions can be chosen at sampling time without re-training, for example emphasizing the text for expressive gestures or the goal for precise navigation.

5. Limitations and Conclusion

Limitations. While our results are promising, several aspects of this work leave room for improvement:

- **Evaluation.** Our evaluation is qualitative only: we do not yet report MPJPE, FID, diversity, or foot-skating metrics against ground truth, nor do we measure error accumulation over long autoregressive rollouts.
- **Data.** NymeriaPlus’s scene annotations are incomplete for some sequences, its captured motion is occasionally imprecise or noisy, and dynamic objects are not represented in the static occupancy grid seen by the scene encoder.
- **Conditions.** The goal is a strong signal relative to text and scene: because it is exactly the pelvis position the window reaches in its last frame, it can dominate the other two at moderate guidance scales instead of being weighed against them as intended. Text, scene, and goal are also not independent in practice, and combining them with large guidance scales can push sampling out of distribution.
- **Compute.** Training and evaluation ran on a single consumer GPU (RTX 3080, 10 GB VRAM), which bounded batch size, model scale, and the amount of hyperparameter search possible within the project’s timeline.

Conclusion. We presented a motion diffusion model that conditions jointly on text, scene, and goal through a shared token sequence, with independent classifier-free guidance for each condition. Trained on real-world data from NymeriaPlus, it produces natural motion on held-out sequences that respects the scene and reaches its goals, and our CFG ablation shows that each guidance weight controls its intended aspect of the motion. More broadly, these results show how a single motion diffusion model can serve many uses, from text-driven animation to scene-aware character control and waypoint navigation.

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